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Chair of Measurement and Sensor Technology

Project Documentation

„Project Lab Embedded Systems “

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Project: Data collection of Electrical Impedance Tomography for phantom validation

Date: 01.07.2025

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1. Abstract

Electrical Impedance Tomography (EIT) is a non-invasive imaging modality that reconstructs the internal conductivity distribution of a subject by injecting small electrical currents through surface electrodes and measuring the resulting voltages. Its advantages include low cost, portability, and the absence of ionizing radiation, making it suitable for continuous monitoring in clinical and industrial environments. EIT relies on solving an inverse problem based on boundary measurements, which is mathematically ill-posed and highly sensitive to noise. The dimensionality of the model—2D or 3D—significantly affects the image quality, computational complexity, and practical applicability of the technique.

2D Electrical Impedance Tomography The 2D EIT approach models the target domain as a planar cross-section, simplifying the forward and inverse problems and reducing computational requirements. It is widely used in applications where the region of interest is approximately uniform in one dimension, such as pipeline monitoring or simplified medical imaging of the thorax. While 2D EIT offers faster image reconstruction and is easier to implement, it suffers from limited depth resolution and can lead to inaccuracies in anatomically complex regions. Despite these limitations, 2D EIT remains valuable for real-time imaging and system prototyping due to its efficiency and robustness in controlled scenarios.

3D Electrical Impedance Tomography 3D EIT extends the technique into volumetric space, enabling more accurate modeling of complex geometries and internal conductivity variations. It is particularly beneficial for clinical applications like brain imaging, lung function monitoring, and breast cancer detection, where anatomical accuracy is crucial. Although 3D EIT requires significantly higher computational power and more sophisticated meshing and inversion algorithms, it offers improved localization and resolution of conductivity changes. The integration of 3D models with advanced regularization methods and parallel computing has enhanced the feasibility of real-time volumetric imaging, marking a critical advancement in the development of EIT as a clinically viable imaging tool.

2. Member Responsibilities

D. Arjun, Sourodipto Bhowmick - Measurement

Remy Lumin Benedict, Charles Mathew - Data analysis

D. Arjun, Sourodipto Bhowmick, Remy Lumin Benedict, Charles Mathew – Documentation

3. Functional Description

3.1 Working Principle of EIT

Electrical Impedance Tomography (EIT) works by injecting small alternating electrical currents into a subject through electrodes placed on its surface and measuring the resulting voltage differences between various pairs of electrodes. The underlying principle is that different tissues (or materials) inside the body (or object) have different electrical conductivities and permittivities. These variations affect how the current flows through the object, which in turn influences the voltages measured at the boundary [8].

The working principle of EIT is divided into two main problems:

Forward Problem: This involves calculating the voltage distribution on the surface of the object when the internal conductivity distribution is known. The object is modeled using a finite element mesh, and the electric potential distribution is computed using Maxwell's equations (usually simplified to the quasi-static approximation of Ohm's and Kirchhoff's laws for low-frequency current). This step is used during simulation and is a critical part of solving the inverse problem.

Inverse Problem: The goal here is to reconstruct the internal conductivity distribution from the measured voltages on the boundary. This is an ill-posed problem because small changes in voltage can correspond to large changes in conductivity, and there is not a unique or stable solution without regularization. Mathematical techniques such as Tikhonov regularization, total variation, or machine learning-based methods are often used to stabilize and solve this problem.

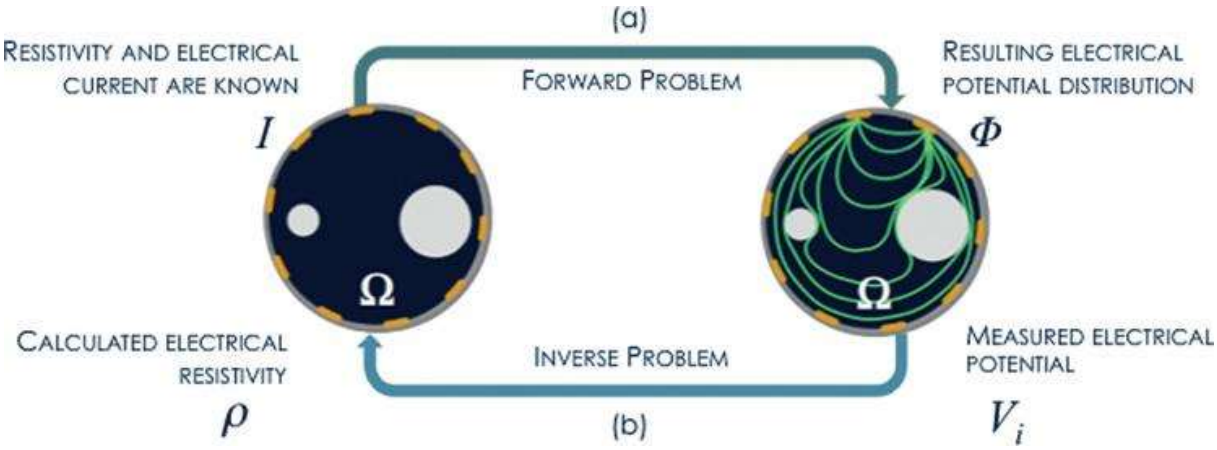

Fig 3.1 Forward and inverse problem [9]

3.2 Current Injection Patterns

A current injection pattern in Electrical Impedance Tomography (EIT) refers to the specific sequence and configuration by which electrical currents are injected into a subject through surface electrodes during data acquisition. EIT works by applying small alternating currents between pairs of electrodes placed on the surface of an object (such as the human body) and measuring the resulting voltages at other electrode locations. The pattern in which these currents are injected—meaning which electrodes are used, in what order, and how often—plays a crucial role in the quality and sensitivity of the reconstructed internal image [3].

- **2D Current Injection Patterns**

In 2D Electrical Impedance Tomography (EIT), the imaging domain is considered as a single, flat cross-section, typically circular or elliptical in shape. Electrodes are placed evenly along the boundary of this plane, forming a ring-like configuration. Current injection in 2D EIT involves selecting a pair of electrodes (injecting and returning) to pass a small alternating current through the body. Voltage measurements are then taken from the remaining electrodes. This process is repeated using multiple electrode pairs to collect a complete set of boundary voltage data. Common current injection patterns in 2D include adjacent, where current is injected between neighboring electrodes; opposite, where the injection pair is placed across the

diameter; and skip or zigzag patterns, where current is injected between electrodes with one or more in between [1].

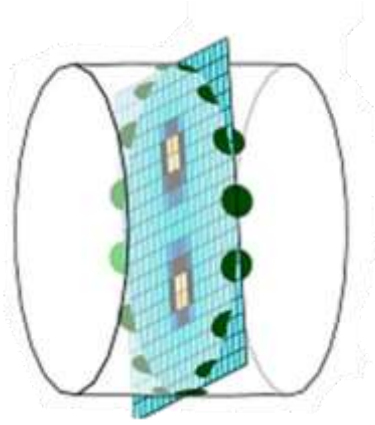

Fig 3.2.1 2D Current injection [11]

- **3D Current Injection Patterns**

3D Electrical Impedance Tomography (EIT) expands upon the 2D approach by considering the full volumetric structure of the object or subject being imaged. In 3D EIT, electrodes are placed not just in a single plane but distributed over the entire surface of the body, often in cylindrical, spherical, or anatomically tailored arrangements. This setup allows for much more diverse current injection patterns, where current can be applied between any two surface electrodes—whether they are located on the same plane, vertically aligned, diagonally positioned, or at varying circumferential and longitudinal locations. These three-dimensional current paths enable deeper penetration of the electric field and improved sensitivity to internal conductivity changes, especially those occurring away from the surface [2].

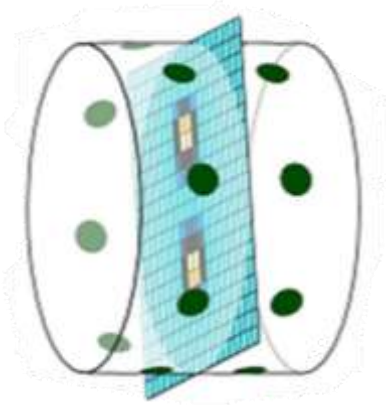

Fig 3.2.2 3D Current injection [11]

3.3 Phantoms

In Electrical Impedance Tomography (EIT), a phantom is a crucial tool used to simulate the electrical properties of biological tissues or other conductive media for the purpose of developing, testing, calibrating, and validating EIT systems. Phantoms allow researchers to

perform controlled experiments without the ethical and logistical challenges of using human or animal subjects. They replicate the geometry, electrical conductivity, and sometimes even the anatomical structure of the body or specific organs, providing a reliable and repeatable environment for imaging tests. There are several types of phantoms used in EIT, each tailored to specific applications and testing needs. One of the most common types is the saline tank phantom, which consists of a transparent container—typically circular or cylindrical—filled with saline solution. Electrodes are attached to the container's inner wall, and objects with known electrical properties (such as plastic, sponge, or metal) are inserted to simulate inclusions or abnormalities. This type of phantom is ideal for basic 2D testing and algorithm validation due to its simplicity and low cost [5].

Table 3.3.1 Types of Phantoms

Phantom Type	Physical/Virtual	Realism	Common Use
Saline Tank	Physical	Low	Basic 2D testing
Agar/Gel	Physical	Medium	2D/3D imaging tests
Anthropomorphic	Physical	High	Clinical validation
3D-Printed	Physical	Medium	Precision & repeatability
Numerical (Digital)	Virtual	Variable	Algorithm development

3.4 Other Components

- **STM32**

Signal Control: STM32 controls the timing and switching of current injection and voltage measurement cycles. Multiplexer Control: It operates analog or digital multiplexers to route signals from/to multiple electrodes. Data Acquisition: STM32 can interface with ADCs (Analog-to-Digital Converters) to sample the voltage data from electrodes.

- **UART**

UART (Universal Asynchronous Receiver/Transmitter) plays a crucial role in EIT data acquisition by enabling serial communication between the EIT hardware and a computer or microcontroller. It transfers digitized impedance data reliably and efficiently, facilitating real-time monitoring, control, and processing of EIT signals during imaging procedures.

- **EVAL-CN0304**

The EVAL-CN0304-SDZ is an evaluation board from Analog Devices designed for the AD9834, a 75 MHz low-power Direct Digital Synthesizer (DDS). This board enables the generation of high-quality sine, triangle, and square waveforms with low power consumption (20 mW at 3 V). It interfaces with a PC via the EVAL-SDP-CB1Z System Demonstration Platform, allowing for USB-to-SPI communication.

4. Experiment

4.1 Procedure

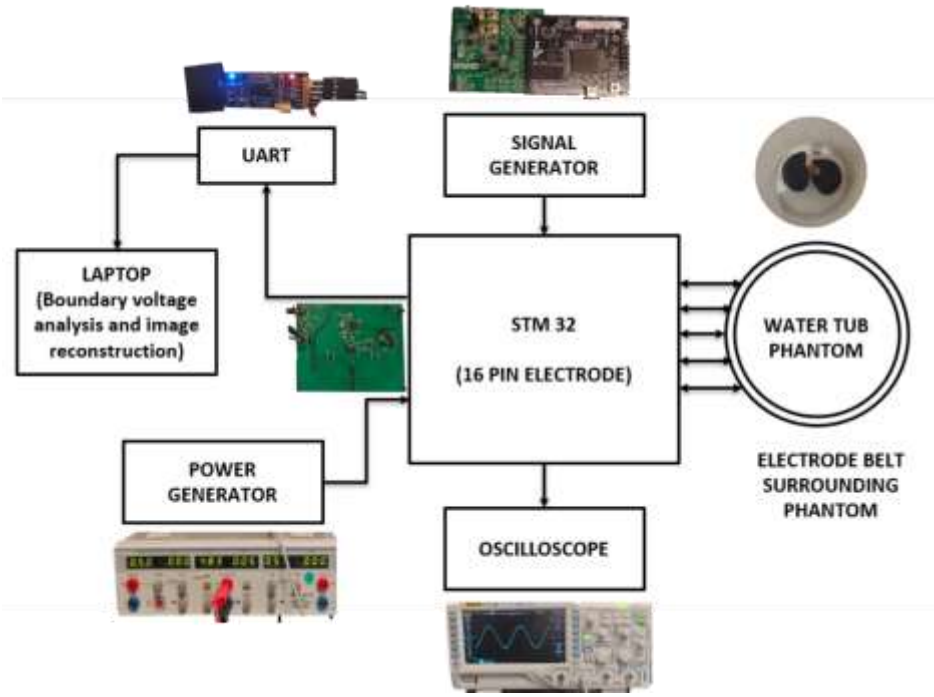

Fig 4.1.1 Flow diagram of EIT

The image presents a comprehensive flowchart demonstrating a signal acquisition and processing system using an STM32 microcontroller. The process begins with a laptop connected to a signal generator, which produces specific test signals. These signals are transmitted directly to the STM32 microcontroller to serve as input for processing. To ensure that the correct signal is received, an oscilloscope is attached to monitor and confirm the signal waveform and integrity before it is processed by the STM32. The microcontroller is powered externally to maintain stability during operation. Connected to the STM32 is a 16-pin interface, which links to an array of electrode commonly used for biological signal acquisition or stimulation purposes in biomedical and neuroscience experiments. Once the signal is received and processed by the STM32 using FFT, the output data is transmitted back to the laptop using UART (Universal Asynchronous Receiver-Transmitter) communication. This UART interface enables efficient, real-time data transfer, which is essential for analysis, visualization, or logging. The entire system is designed for precision signal testing, validation, and data acquisition applications, particularly in embedded systems or bio-signal research. This setup enables developers and researchers to prototype and analyze custom signal processing workflows in a controlled and observable manner [4].

- Different connection of electrodes

Fig 4.1.2 (a) 2D Electrode Placement (b) 3D Electrode Placement

- Different experiment methodology

Fig 4.1.3 (a) Water (b) Lung

Fig 4.1.4 (a) Plastic Inside (b) Plastic Outside

Fig 4.1.5 (a) Iron Inside (b) Iron Outside

Fig 4.1.6 (a) Plastic Inside Iron Outside (b) Iron Inside Plastic Outside

4.2 Software

- **RealTerm:**

RealTerm is a powerful serial terminal software used for capturing and analyzing serial data. It serves mainly as a low-level communication and data logging tool.

1. Connect STM32 to the PC

- Use a USB to UART bridge
- Confirm the correct COM port in Device Manager

2. Open RealTerm and Configure It

2.1. Port tab

- Choose the correct COM port
- Set Baud rate, Data bits (8), Parity (None), Stop bits (1)
- Click “Open” to begin communication

2.2. Display tab

- Set to ASCII, Hex+ASCII, or Hex depending on your data format
- For human-readable EIT data, use ASCII

2.3. Capture tab

- Choose a file location and click Start: Capture to File
- RealTerm will now log incoming UART data to a file

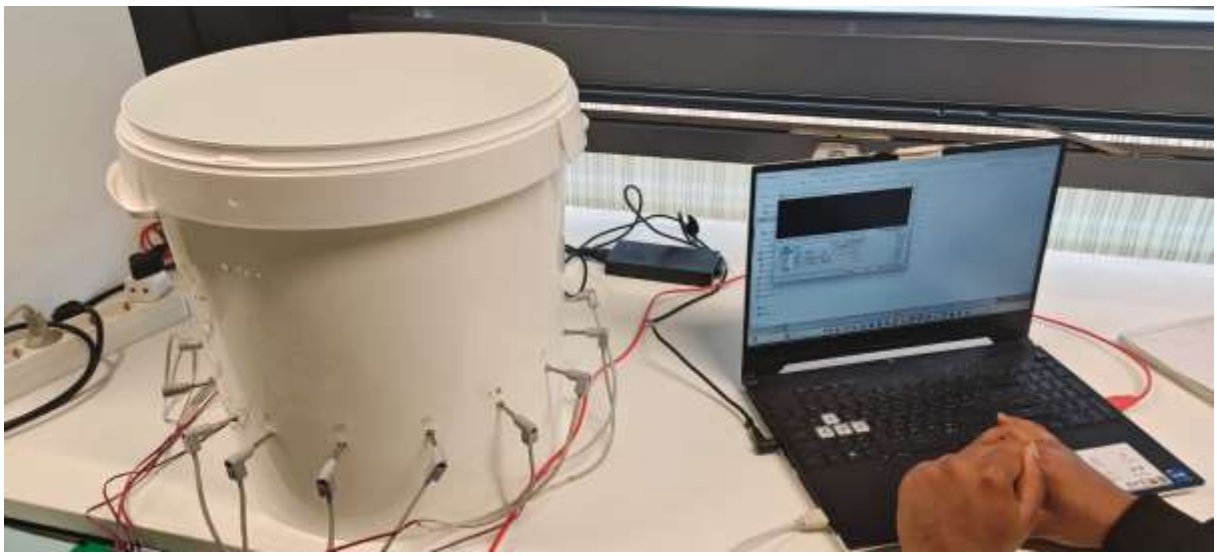

Fig 4.2.1 Measurement using RealTerm

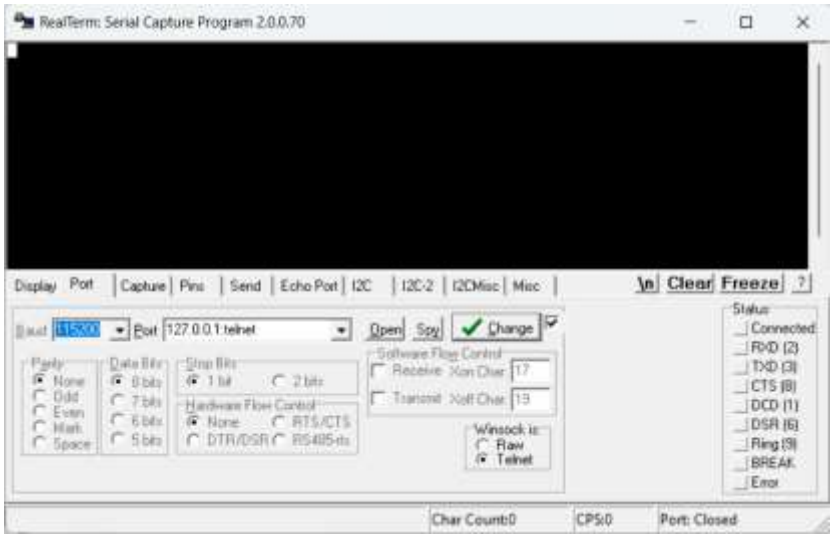

Fig 4.2.2 RealTerm Software

- **Analog Devices EVAL-CN0304 Software:**

1. Connect the hardware

   - Connect the **SDP-CB1Z** board to your **PC via USB**
   - Connect the **evaluation board** (EVAL-CN0304-SDZ) to the **SDP-CB1Z** using the provided connector (typically via 120-pin interface)
   - Power on the eval board (if external supply is needed)

2. Launch the software

   - Go to the Start Menu or your Desktop and find **AD9834 Evaluation Software**

3. Launch the application

- The software will attempt to **detect the SDP-CB1Z** and **the connected board**

4. Use the GUI

- Set frequency, phase, output mode (sine, triangle, square), and sweep parameters
- Use buttons to:
  - Write to FREQ/PHASE registers
  - Enable/disable DAC or Comparator
  - Apply RESET or SLEEP modes

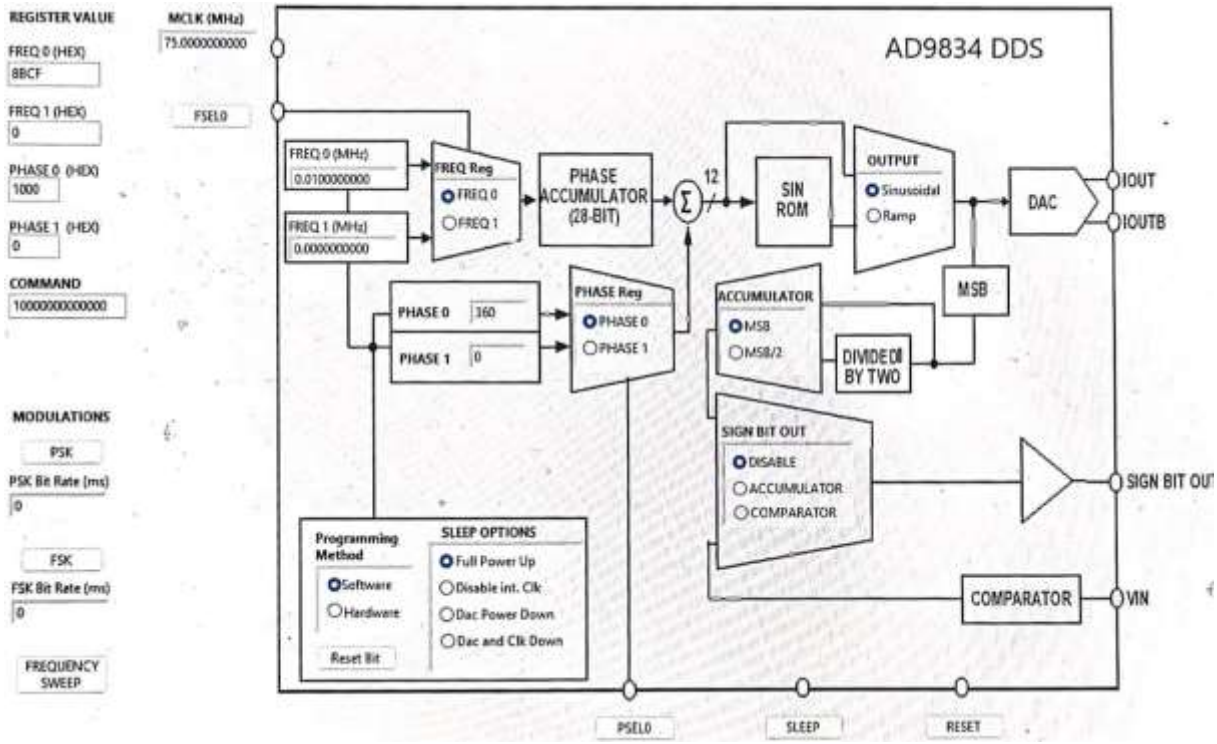

Fig 4.2.3 Signal generation using EVAL-CN0304

5. Empirical Data and Analysis

The measured voltage values are saved in .txt Format using RealTerm software and they are transferred to Excel. These values are later used for data analysis.

For 2D - Vertical, Opposite (Relative), Adjacent, Rotating Radial (New)

For 3D - Planar, Square, ZigZag

1. Lung
2. water
3. Iron inside
4. Iron outside
5. Plastic inside

6. Plastic outside
7. Iron inside Plastic outside
8. Plastic inside Iron outside

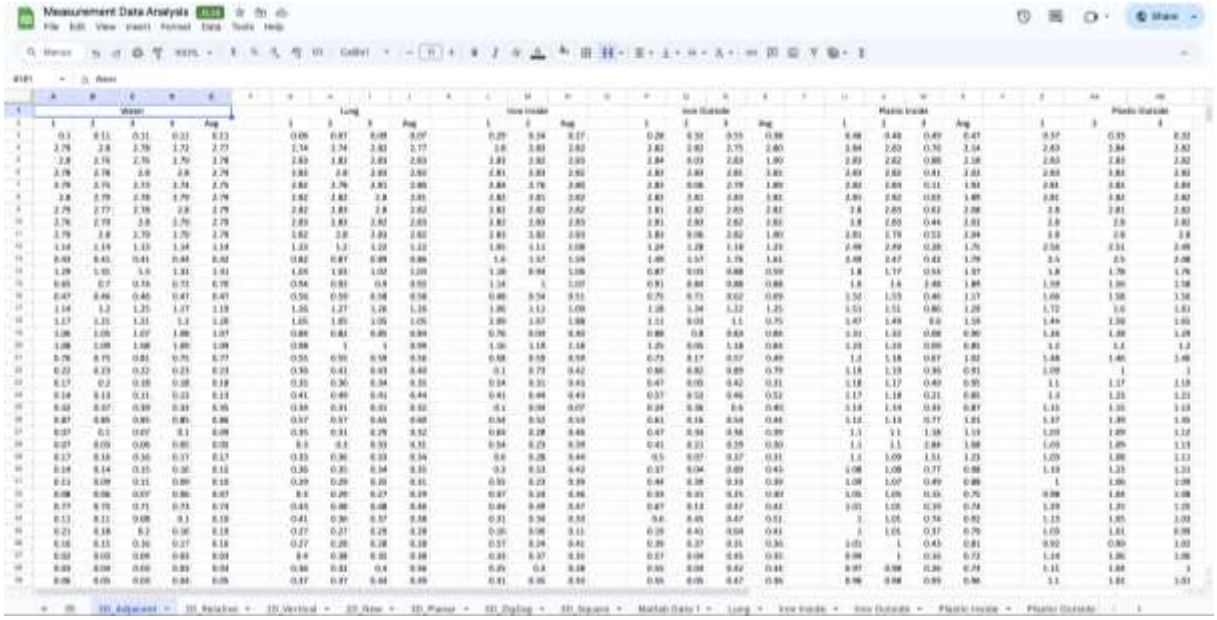

Fig 5.1 Excel measurement data

In this study, data analysis was conducted on Electrical Impedance Tomography (EIT) measurements using Microsoft Excel. The raw voltage or impedance data collected from multiple electrode pairs was first organized in Excel for clarity. Data cleaning was performed to remove any anomalies or missing values. Following this, the dataset was segmented according to number of electrodes (208 for Adjacent, 192 for Opposite and Vertical, 340 for Rotating Radial) and voltage values comparative analysis.

Graphs were created to visualize trends and patterns in the voltage values. Line graphs were used to plot voltage versus electrode pair combinations for monitoring dynamic changes. Bar graphs and scatter plots were also employed to highlight specific variations or outliers. These visualizations helped identify consistent signal responses and detect areas of abnormal impedance that could correspond to conductivity changes within the medium. Overall, Excel provided a flexible and accessible environment for performing preliminary EIT data processing, enabling effective visualization and interpretation of the complex dataset [8].

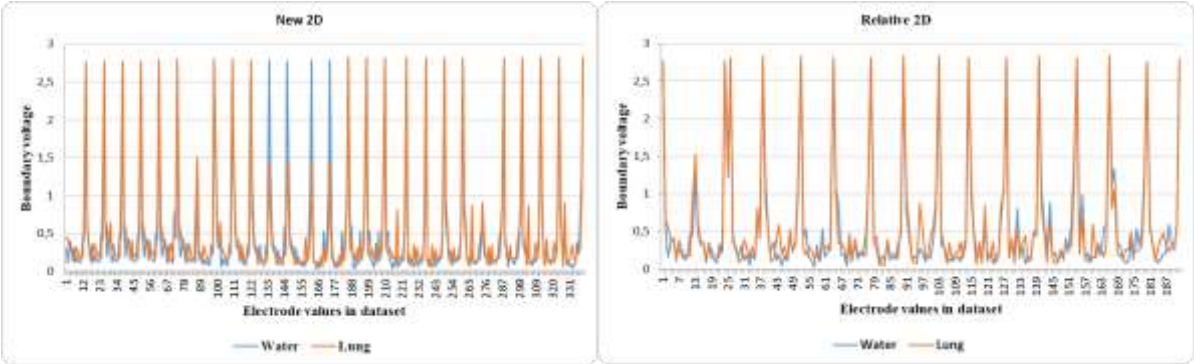

Fig 5.2 Comparison of Rotating Radial (New) and Opposite (Relative) method using Water and Lung

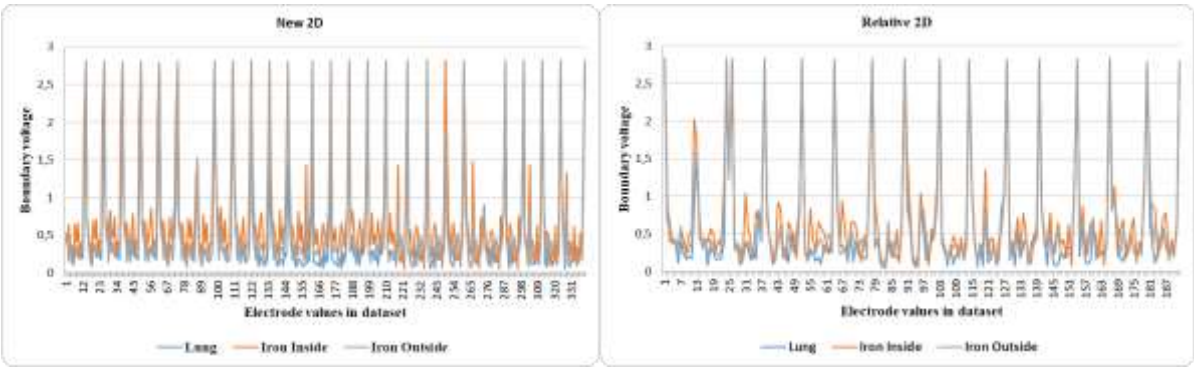

Fig 5.3 Comparison of Rotating Radial (New) and Opposite (Relative) method using Lung, Iron Inside and Iron Outside

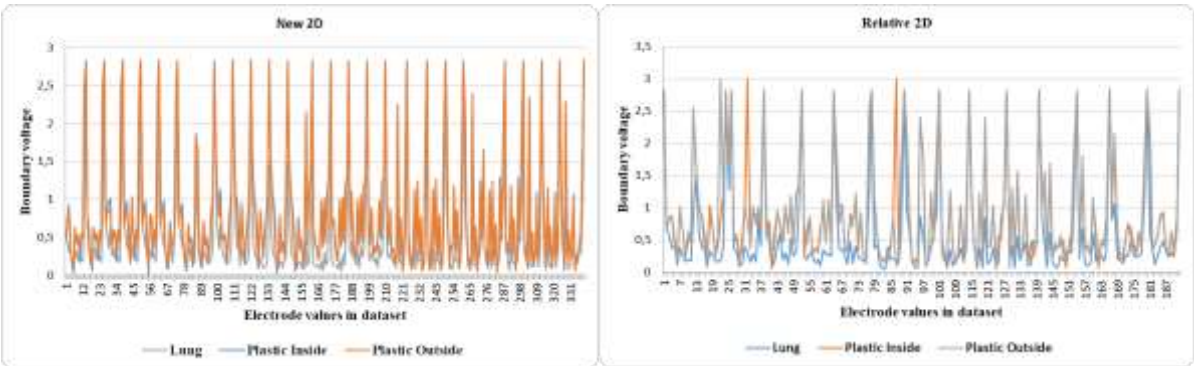

Fig 5.4 Comparison of Rotating Radial (New) and Opposite (Relative) method using Lung, Plastic Inside and Plastic Outside

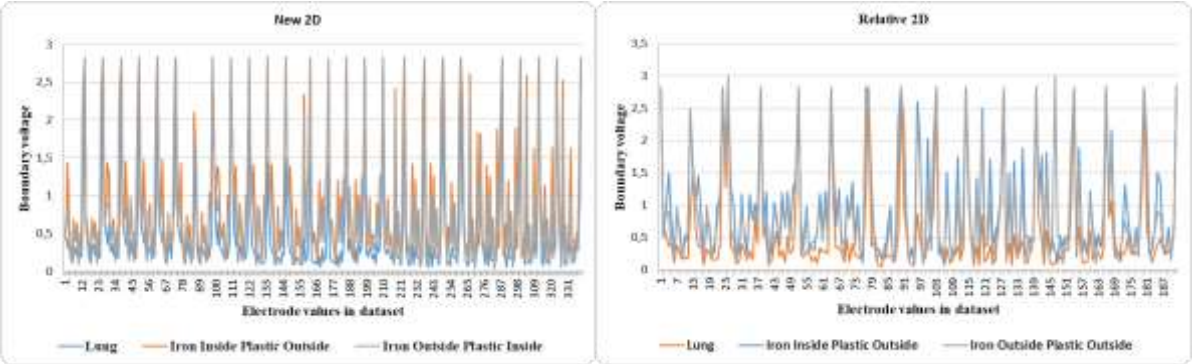

Fig 5.5 Comparison of Rotating Radial (New) and Opposite (Relative) method using Lung, Iron Inside Plastic Outside and Iron Outside Plastic Inside

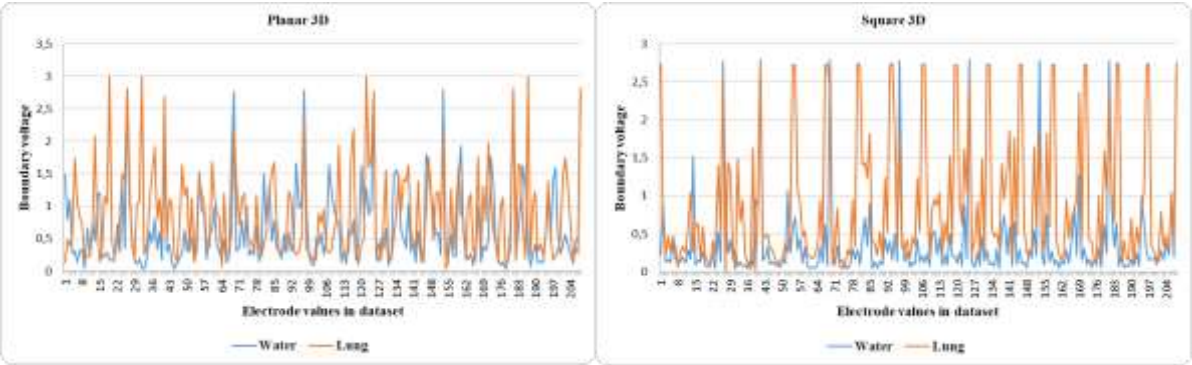

Fig 5.6 Comparison of Planar and Square method using Water and Lung

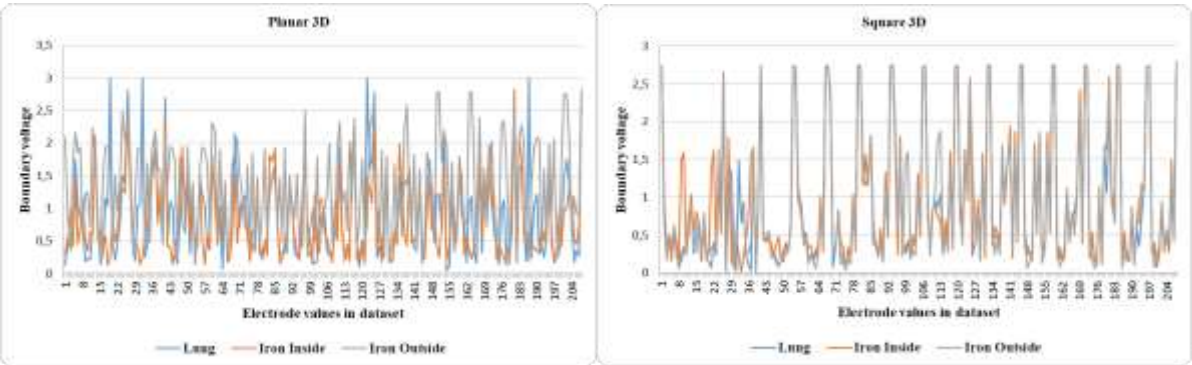

Fig 5.7 Comparison of Planar and Square method using Lung, Iron Inside and Iron Outside

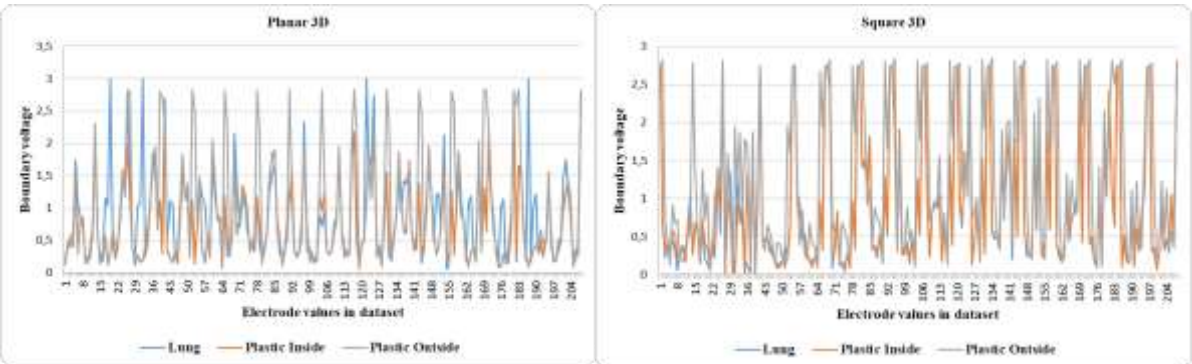

Fig 5.8 Comparison of Planar and Square method using Lung, Plastic Inside and Plastic Outside

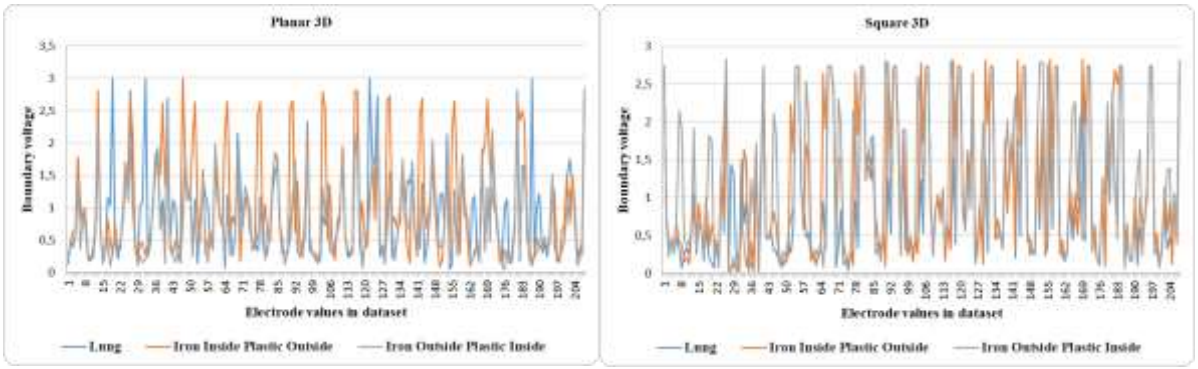

Fig 5.9 Comparison of Planar and Square method using Lung, Iron Inside Plastic Outside and Iron Outside Plastic Inside

6. Image Reconstruction

The Gauss-Newton algorithm is a widely used method for solving nonlinear least-squares problems and plays a crucial role in Electrical Impedance Tomography (EIT) for reconstructing internal conductivity distributions. In the pyEIT library, the Gauss-Newton method is implemented to solve the inverse problem by iteratively minimizing the difference between measured and simulated voltage data on the boundary of an object. To use it, one begins by creating a finite element mesh of the imaging domain and defining a stimulation and measurement protocol. The forward problem is solved for a baseline (homogeneous) case and again after introducing anomalies (inhomogeneous case). These simulated voltages are then used as input to the Gauss-Newton solver. The GN class in pyEIT is initialized with parameters like the regularization factor (lamb) and prior weighting (p), helping stabilize the solution. After setting up, the solver computes the conductivity changes using the difference imaging technique. The result is a set of conductivity values associated with mesh elements. These values can be interpolated over a grid and visualized to show the location and magnitude of conductivity changes within the object. The Gauss-Newton method in pyEIT is effective, especially when high-resolution reconstruction is needed in static or time-difference EIT.

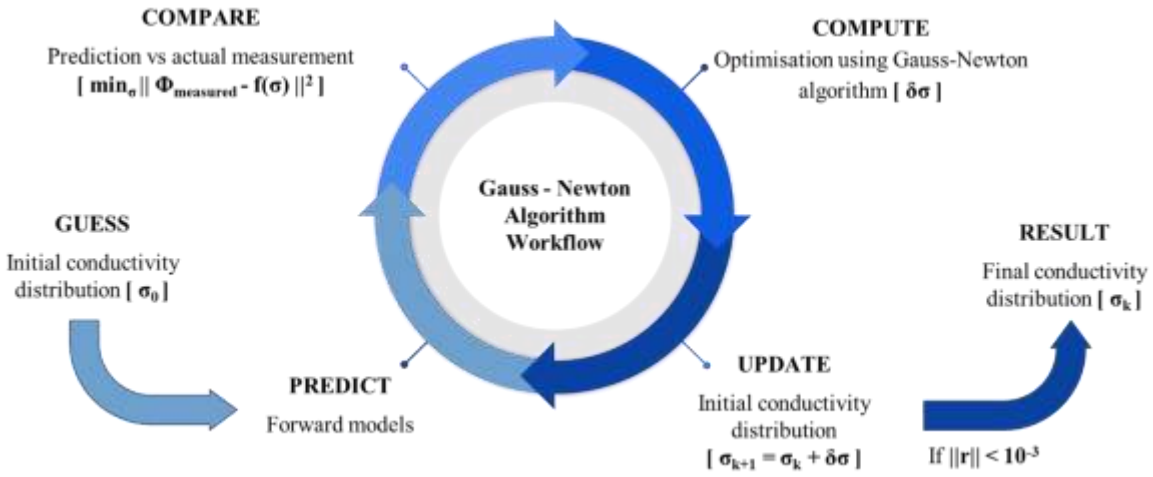

Fig 6.1 Gauss – Newton algorithm workflow

The Gauss-Newton algorithm in pyEIT provides an estimate of internal conductivity changes by minimizing the difference between measured and simulated voltages. To evaluate the quality of these reconstructions, Root Mean Square (RMS) error and the Correlation Coefficient (CC) are commonly used as quantitative metrics. The RMS error measures the average magnitude of reconstruction errors. It quantifies how close the reconstructed conductivity values are to the true or expected values. A lower RMS error indicates a more accurate reconstruction produced by the Gauss-Newton algorithm. The Correlation Coefficient assesses the linear relationship between the reconstructed image and the ground truth distribution. A value close to 1 implies a strong positive correlation, meaning the Gauss-Newton method has correctly captured the spatial pattern of conductivity changes. But we couldn't finish the image reconstruction for all cases since it's highly computational consuming and results are still running.

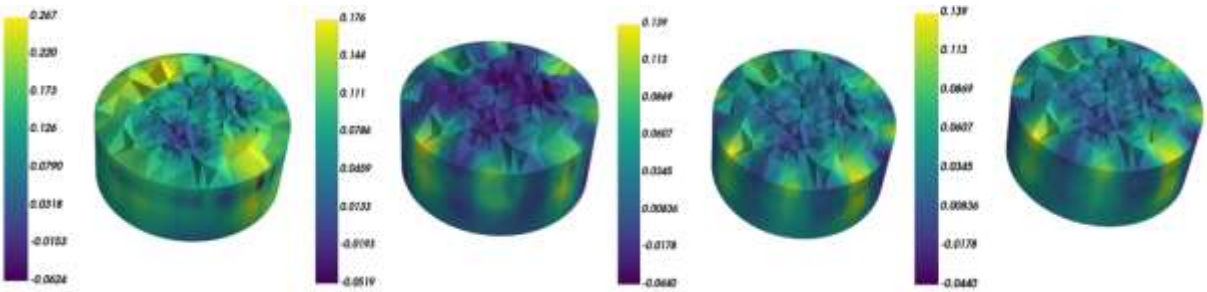

Fig 6.2 Image reconstruction of *Lung in Water* by 2D Adjacent, Vertical, Opposite, Rotating Radial

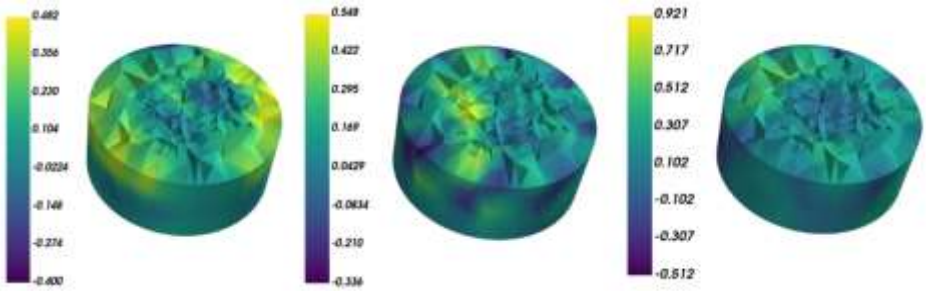

Fig 6.3 Image reconstruction of *Lung in Water* by 3D Planar, ZigZag, Square

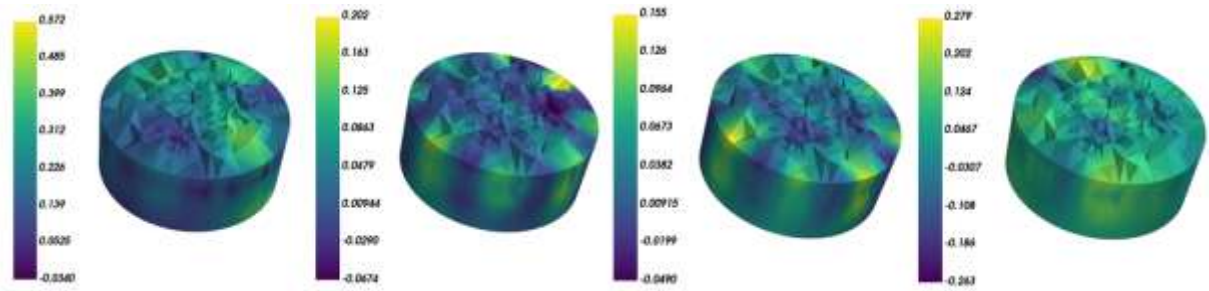

Fig 6.4 Image reconstruction of *Iron Inside* by 2D Adjacent, Vertical, Opposite, Rotating Radial

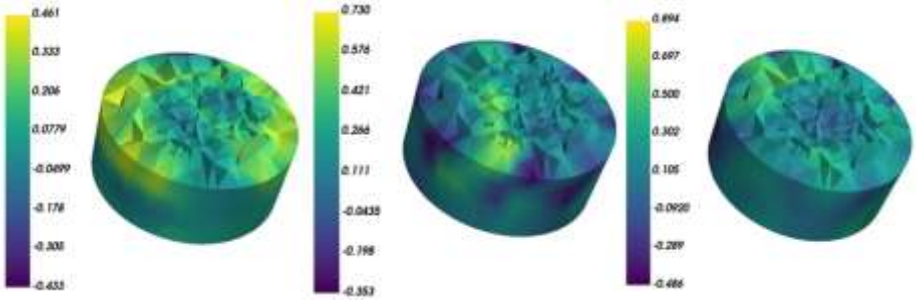

Fig 6.5 Image reconstruction of *Iron Inside* by 3D Planar, ZigZag, Square

7. Results

In Electrical Impedance Tomography (EIT) measures how well two different conductivity states can be differentiated by calculating the Euclidean (L2) distance between their measured voltage or impedance data vectors. A higher value means the two states are more easily distinguishable. This metric helps evaluate the sensitivity and accuracy of the EIT system, guiding improvements in electrode placement and measurement methods to better detect anomalies like tumors or fluid changes within the body.

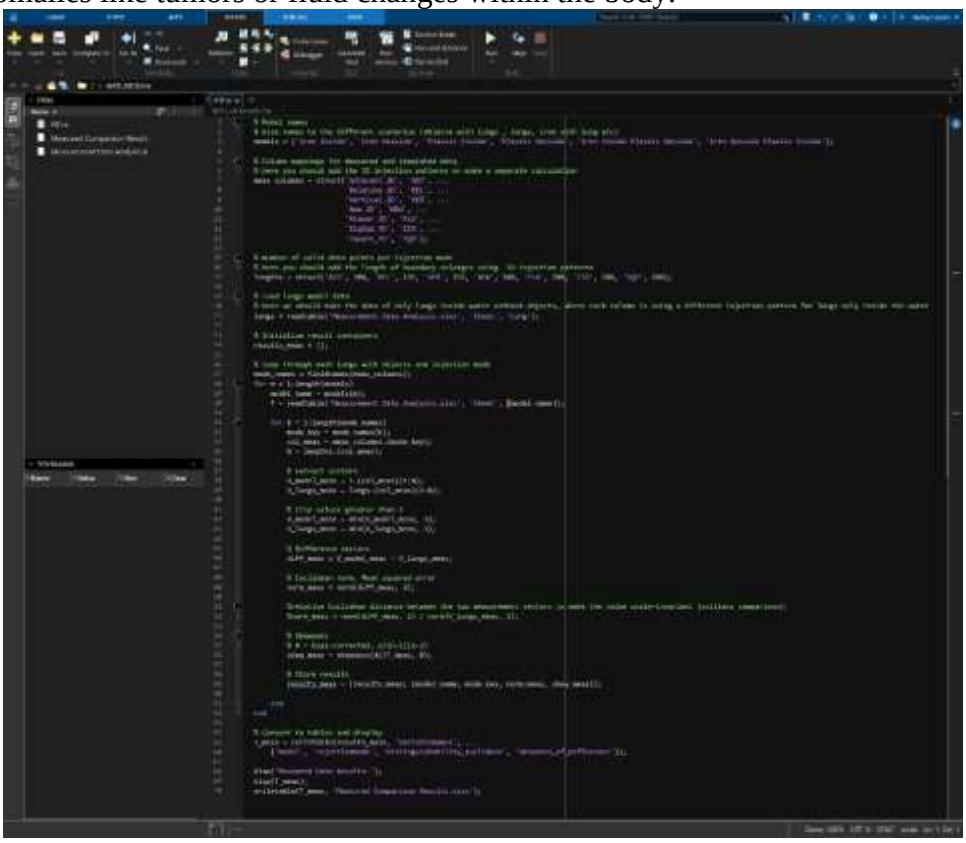

Fig 7.1 MATLAB script

Table 7.1 Results based on distinguishability Euclidean

Model	Injection Mode	Distinguishability Euclidean	Model	Injection Mode	Distinguishability Euclidean

Iron Inside	Adjacent_2D	6,195054479	Iron Outside	Adjacent_2D	3,584954672
	Opposite_2D	3,609348972		Opposite_2D	2,542085758
	Vertical_2D	4,289370583		Vertical_2D	1,789441254
	Rotating Radial_2D	8,706043878		Rotating Radial_2D	6,969002798
	Planar_3D	12,34550525		Planar_3D	15,3651261
	ZigZag_3D	3,259877298		ZigZag_3D	11,1293441
	Square_3D	3,333421665		Square_3D	1,7435309
Plastic Inside	Adjacent_2D	5,481724181	Plastic Outside	Adjacent_2D	7,082040666
	Opposite_2D	15,52651138		Opposite_2D	6,789094196
	Vertical_2D	6,451123933		Vertical_2D	6,511151972
	Rotating Radial_2D	11,45023144		Rotating Radial_2D	11,44699087
	Planar_3D	11,77671856		Planar_3D	14,41978502
	ZigZag_3D	8,962432705		ZigZag_3D	7,659203614
	Square_3D	2,370210961		Square_3D	12,67900233
Iron Inside Plastic Outside	Adjacent_2D	6,387863493	Iron Outside Plastic Inside	Adjacent_2D	5,180801096
	Opposite_2D	8,148104074		Opposite_2D	15,28729865
	Vertical_2D	7,129305717		Vertical_2D	5,857764079
	Rotating Radial_2D	13,87409096		Rotating Radial_2D	11,22825899
	Planar_3D	17,00015		Planar_3D	11,88981917
	ZigZag_3D	7,862645865		ZigZag_3D	11,14777556
	Square_3D	7,641420025		Square_3D	11,85178468

8. Conclusion

In practice, evaluating Electromagnetic Induction Tomography (EIT) data involves several key steps. We begin by assessing distinguishability to ensure the system can detect internal changes, if this metric is too low, abnormality may go unnoticed. Next, we calculate the Euclidean distance to evaluate how closely the reconstructed abnormality aligns. Finally, we examine the Root Mean Square Error (RMSE) to gauge the accuracy of reconstructed conductivity values. Together, these metrics provide insight into the reliability and practical usability of the reconstruction. By the result produced by the MATLAB script, it clearly shows for 2D Rotating Radial (New) is better than Opposite (Relative). For 3D, Planar is the best method.

9. References

[1] Zhang, Lifeng, and Huaxiang Wang. "Single source current drive patterns for electrical impedance tomography." *2010 IEEE Instrumentation & Measurement Technology Conference Proceedings*. IEEE, 2010.

[2] Graham, B. M., and A. Adler. "Electrode placement configurations for 3D EIT." *Physiological measurement* 28.7 (2007): S29.

[3] Schullcke, Benjamin, et al. "Simultaneous application of two independent EIT devices for real-time multi-plane imaging." *Physiological Measurement* 37.9 (2016): 1541.

[4] Hafsa, Mariem, et al. "3d image reconstruction based on electrical impedance tomography measurements using a gauss-newton algorithm." *2021 International Workshop on Impedance Spectroscopy (IWIS)*. IEEE, 2021.

[5] Shi, Yanyan, et al. "Information Enhancement With Multilayer Convolutional Neural Network for Accurate Lung Imaging." *IEEE Internet of Things Journal* (2024).

[6] Menden, T., et al. "Development of an electrical phantom for multi-frequency electrical impedance tomography based on the visible human project." *International Journal of Bioelectromagnetism* (2018): 51.

[7] Harrach, Bastian, Jin Keun Seo, and Eung Je Woo. "Factorization method and its physical justification in frequency-difference electrical impedance tomography." *IEEE Transactions on Medical Imaging* 29.11 (2010): 1918-1926.

[8] Bader, Oumaima, Najoua Essoukri Ben Amara, and Olfa Kanoun. *Evaluation of 3d current injection patterns for human lung monitoring in electrical impedance tomography*. TU Ilmenau, 2023.

[9] Faia, P. M., et al. "Imaging particulate two-phase flow in liquid suspensions with electric impedance tomography." *Particulate Science and Technology* 30.4 (2012): 329-342.

[10] Mansouri, Sofiene, et al. "Electrical Impedance tomography–recent applications and developments." *Journal of electrical bioimpedance* 12.1 (2021): 50.

[11] Brabant, Olivia A., et al. "Thoracic electrical impedance tomography—the 2022 veterinary consensus statement." *Frontiers in veterinary science* 9 (2022): 946911.